Instituto tecnológico de Querétaro

Conm. y Enrut. Redes de Datos

Teacher: Diaz Sánchez Ricardo

Student Name: Martinez Pérez José Armando

Group: 6A

Major: Ing. Sistemas computacionales

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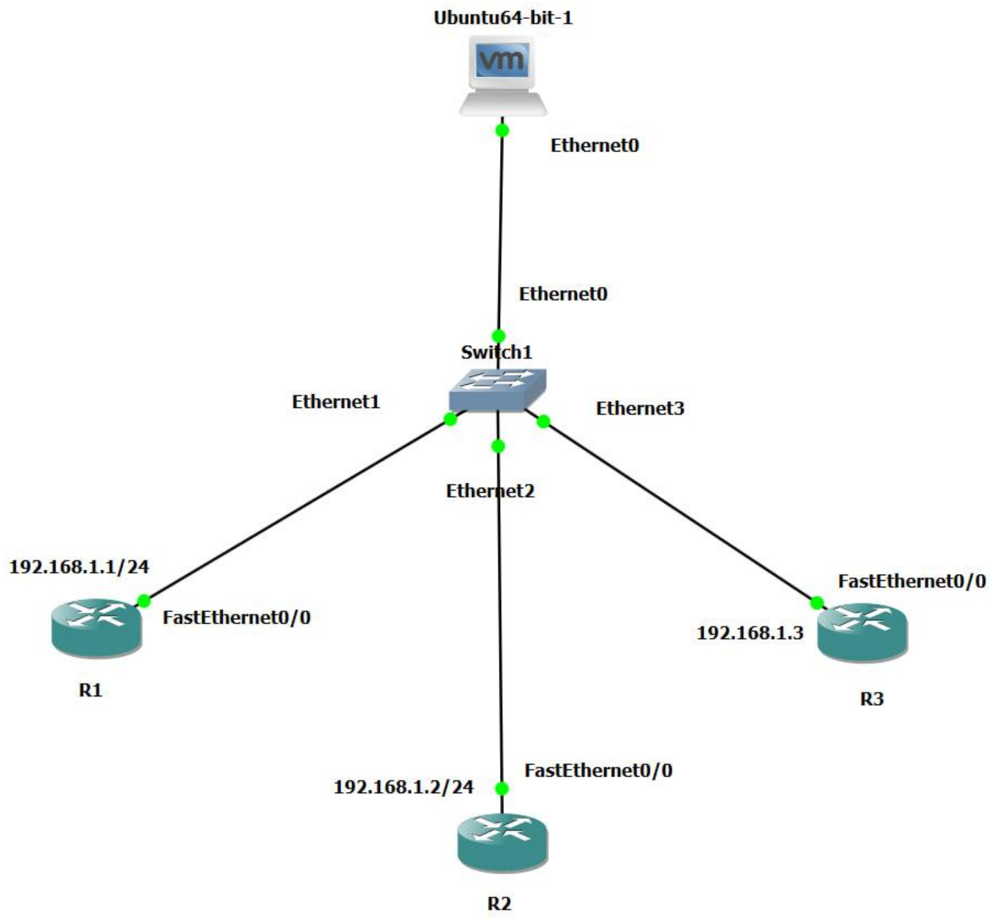
Activity: Nornir with netmiko

Step by Step to configure this activity.

1. Create the topology in GNS3
    a. Ubuntu on VMware
    b. Switch Ethernet
    c. x3 Cisco Routers



2. Configure the routers:

   a. Addressing information

```
R1(config)#int f0/0
R1(config-if)#ip add 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#
*May  6 17:28:40.515: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*May  6 17:28:41.515: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
```

```
R2(config)#int f0/0
R2(config-if)#ip add 192.168.1.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#
*May  6 17:30:02.991: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*May  6 17:30:03.991: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R2(config)#
```

```
R3(config)#int f0/0
R3(config-if)#ip add 192.168.1.3 255.255.255.0
R3(config-if)#
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*May  6 17:31:10.139: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*May  6 17:31:11.139: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R3(config)#
```

b. Configure SSH on each router

```
R1(config)#ip domain-name itq.edu.mx
R1(config)#username admin privilege 15 secret cisco
R1(config)#crypto key generate rsa modulus 1014
The name for the keys will be: R1.itq.edu.mx

% The key modulus size is 1014 bits
% Generating 1014 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 1 seconds)

R1(config)#
*May  6 17:33:14.543: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#crypto key generate rsa modulus 1024
% You already have RSA keys defined named R1.itq.edu.mx.
% They will be replaced.

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 1 seconds)

R1(config)#
*May  6 17:33:19.239: %SSH-5-DISABLED: SSH 1.99 has been disabled
*May  6 17:33:20.147: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#
R1(config)#enable secret cisco
R1(config)#
```

```
R2(config)#ip domain-name itq.edu.mx
R2(config)#username admin privilege 15 secret cisco
R2(config)#crypto key generate rsa modulus 1024
The name for the keys will be: R2.itq.edu.mx

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 0 seconds)

R2(config)#
*May  6 17:35:08.783: %SSH-5-ENABLED: SSH 1.99 has been enabled
R2(config)#
R2(config)#line vty 0 4
R2(config-line)#login local
R2(config-line)#transport input ssh
R2(config-line)#exit
R2(config)#
R2(config)#enable secret cisco
R2(config)#
```

```
R3(config)#ip domain-name itq.edu.mx
R3(config)#username admin privilege 15 secret cisco
R3(config)#crypto key generate rsa modulus 1024
The name for the keys will be: R3.itq.edu.mx

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 1 seconds)

R3(config)#
R3(config)#
*May  6 17:36:54.607: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exit
R3(config)#
R3(config)#enable secret cisco
R3(config)#
```

3. Update the operating system of the Ubuntu.

```
ubuntu@ubuntu:~/Desktop$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [412 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,662 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [262 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.9 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [13.4 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [20.0 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.0 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [799 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,184 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,962 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [227 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe Icons (48x48) [49.8 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe Icons (64x64) [78.3 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [23.1 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted i386 Packages [25.4 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,929 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [613 kB]
```

4. Installation of python3

```
ubuntu@ubuntu:~/Desktop$ sudo apt install python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential cpp-13 cpp-13-x86-64-linux-gnu dpkg dpkg-dev
  fakeroot g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-14-base gcc-x86-64-linux-gnu javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libasan8 libatomic1 libbinutils libcc1-0 libctf-nobfd0 libctf0 libdpkg-perl libexpat1
  libexpat1-dev libfakeroot libfile-fcntllock-perl libgcc-13-dev libgcc-s1 libgomp1 libgprofng0 libhwasan0 libitm1
  libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0 libpython3-dev libpython3-stdlib libpython3.12-dev
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libquadmath0 libsframe1 libstdc++-13-dev libstdc++6
  libtsan2 libubsan1 lto-disabled-list make python3 python3-dev python3-minimal python3-pkg-resources
  python3-setuptools python3-wheel python3.12 python3.12-dev python3.12-minimal zlib1g-dev
Suggested packages:
  binutils-doc gprofng-gui gcc-13-locales cpp-13-doc debsig-verify debian-keyring g++-multilib g++-13-multilib
  gcc-13-doc gcc-multilib autoconf automake libtool flex bison gcc-doc gcc-13-multilib gdb-x86-64-linux-gnu apache2
  | lighttpd | httpd git bzr libstdc++-13-doc make-doc python3-doc python3-tk python3-venv python-setuptools-doc
  python3.12-venv python3.12-doc binfmt-support
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev fakeroot g++ g++-13
```

5. Install the libraries to use in this practice

```
ubuntu@ubuntu:~/Desktop$ pip3 install nornir nornir_netmiko nornir_utils --break--system-packages

Usage:
  pip3 install [options] <requirement specifier> [package-index-options] ...
  pip3 install [options] -r <requirements file> [package-index-options] ...
  pip3 install [options] [-e] <vcs project url> ...
  pip3 install [options] [-e] <local project path> ...
  pip3 install [options] <archive url/path> ...

no such option: --break--system-packages
ubuntu@ubuntu:~/Desktop$ pip3 install nornir nornir_netmiko nornir_utils --break-system-packages
Defaulting to user installation because normal site-packages is not writeable
Collecting nornir
  Downloading nornir-3.5.0-py3-none-any.whl.metadata (5.4 kB)
Collecting nornir_netmiko
  Downloading nornir_netmiko-1.0.1-py3-none-any.whl.metadata (1.0 kB)
Collecting nornir_utils
```

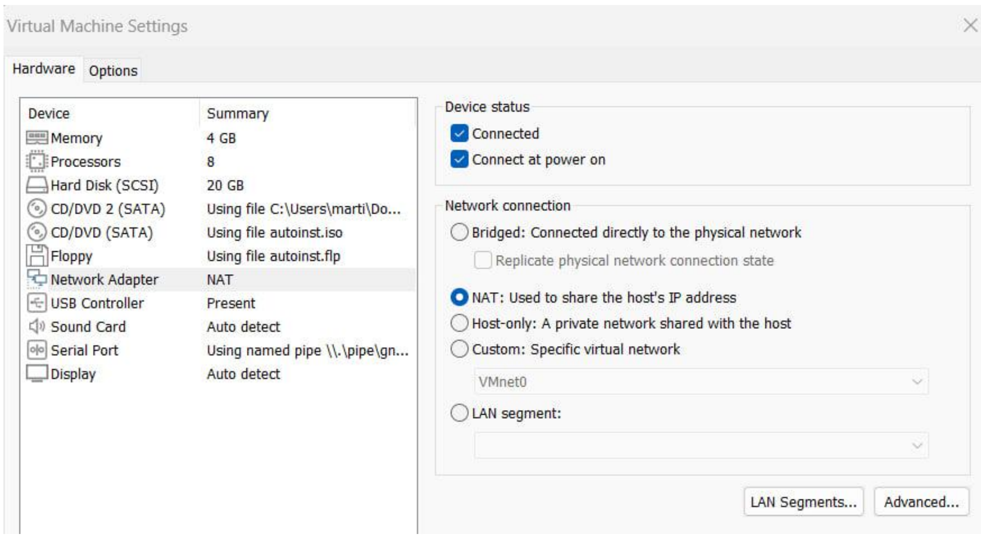
6. Verify the correct libraries installation

```
ubuntu@ubuntu:~/Desktop$ python3 -c "import nornir; import nornir_netmiko; print('librerias listas')"
librerias listas
ubuntu@ubuntu:~/Desktop$
```

7.Configure the IP at the interface on the Ubuntu VM.

```
ubuntu@ubuntu:~/Desktop$ sudo ip addr flush dev ens33
ubuntu@ubuntu:~/Desktop$ sudo ip addr add 192.168.1.100/24 dev ens33
ubuntu@ubuntu:~/Desktop$ sudo ip link set ens33 up
```

8. Configure the network adapter in the Ubuntu VM as Host-Only



9. Create the folder in the VM and work into it

```
ubuntu@ubuntu:~/Desktop$ mkdir -p ~/Desktop/nornir_itq
ubuntu@ubuntu:~/Desktop$ cd ~/Desktop/nornir_itq
ubuntu@ubuntu:~/Desktop/nornir_itq$
```

10. Create the configuration files

- Hosts.yaml

```
ubuntu@ubuntu:~/Desktop/nornir_itq$
ubuntu@ubuntu:~/Desktop/nornir_itq$ nano hosts.yaml
```

```
  GNU nano 7.2                    hosts.yaml *
R1:
  hostname: 192.168.1.1
  groups: [ios]
R2:
  hostname: 192.168.1.2
  groups: [ios]
R3:
  hostname: 192.168.1.3
  groups: [ios]
```

- Groups.yaml

```
ubuntu@ubuntu:~/Desktop/nornir_itq$ nano groups.yaml
ubuntu@ubuntu:~/Desktop/nornir_itq$
```

```
  GNU nano 7.2                          groups.yaml
ios:
  platform: ios
```

- Default.yaml

```
ubuntu@ubuntu:~/Desktop/nornir_itq$ nano defaults.yaml
ubuntu@ubuntu:~/Desktop/nornir_itq$
```

```
  GNU nano 7.2                          defaults.yaml
username: admin
password: cisco
```

- Config.yaml

```
ubuntu@ubuntu:~/Desktop/nornir_itq$ nano config.yaml
ubuntu@ubuntu:~/Desktop/nornir_itq$
```

```
  GNU nano 7.2                          config.yaml
inventory:
  plugin: SimpleInventory
  options:
    host_file: "hosts.yaml"
    group_file: "groups.yaml"
    defaults_file: "defaults.yaml"
runner:
  plugin: threaded
  options:
    num_workers: 10
```

11. Verify the file in the directory

```
ubuntu@ubuntu:~/Desktop/nornir_itq$ ls
config.yaml  defaults.yaml  groups.yaml  hosts.yaml  main.py
ubuntu@ubuntu:~/Desktop/nornir_itq$
```

12. Create the script in py

```
ubuntu@ubuntu:~/Desktop/nornir_itq$ nano main.py
ubuntu@ubuntu:~/Desktop/nornir_itq$
```

```
  GNU nano 7.2                                  main.py
from nornir import InitNornir
from nornir_netmiko.tasks import netmiko_send_command
from nornir_utils.plugins.functions import print_result

# Cargamos la configuracion
nr = InitNornir(config_file="config.yaml")

# Ejecutamos el comando en los 3 routers
resultado = nr.run(
        task=netmiko_send_command,
        command_string="show ip interface brief"
)

# Mostramos el resultado en la terminal
print_result(resultado)
```

12. Execute the script on the Ubuntu VM

```
ubuntu@ubuntu:~/Desktop/nornir_itq$ python3 main.py
netmiko_send_command****************************************************************
* R1 ** changed : False ************************************************************
vvvv netmiko_send_command ** changed : False vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv INFO
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            192.168.1.1     YES manual up                    up
^^^^ END netmiko_send_command ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
* R2 ** changed : False ************************************************************
vvvv netmiko_send_command ** changed : False vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv INFO
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            192.168.1.2     YES manual up                    up
^^^^ END netmiko_send_command ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
* R3 ** changed : False ************************************************************
vvvv netmiko_send_command ** changed : False vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv INFO
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            192.168.1.3     YES manual up                    up
^^^^ END netmiko_send_command ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

Conclusion

This was useful to understand two different technologies and how to combine them to use all the powerful provided by these technologies. Furthermore, it also demonstrates the power of the GNS3 software where the devices can run and use real ISO of the network devices, it avoids the restrictions that some networks devices in another software have. GNS3 is the perfect software to work with devices as if they were real network devices. The possibilities are endless and the limit is in your imagination.

The practice facilitates understanding of how nornir can access to different devices simultaneously and how netmiko can automatically execute command in another network device through remote connection managed by SSH.